

BASE CARLINI, Antarctica

WMO: 890530

Lat: 62.238S

Lon: 58.665W

Elev: 11

StdP: 101.19

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-19.0	-16.2	-22.1	0.5	-18.6	-19.4	0.7	-15.9	28.4	-0.2	25.0	-4.0	9.9	50	1.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	6.1	4.4	5.2	3.8	4.7	3.3	4.7	5.7	4.1	4.9	3.6	4.4	9.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.7	4.9	4.9	3.2	4.8	4.3	2.7	4.6	3.9	18.4	5.9	17.2	5.1	16.3	4.5	8.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.9	21.4	19.7	DB	-20.5	9.6	3.3	2.3	-22.9	11.3	-24.8	12.6	-26.7	13.9	-29.1	15.6	(4)
(5)			WB	-20.9	6.6	3.1	1.1	-23.1	7.4	-24.9	8.1	-26.6	8.7	-28.9	9.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	2.2	2.1	1.1	-1.4	-2.4	-4.9	-5.8	-5.6	-4.0	-1.7	0.0	1.2		
	DBStd	4.40	1.42	1.69	2.37	3.47	3.46	4.51	4.99	4.82	4.24	2.77	2.04	1.49		
	HDD10.0	4244	241	222	277	343	383	447	491	483	421	363	301	272		
	HDD18.3	7286	499	455	535	593	642	697	749	742	671	621	551	530		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.6	8.0	8.8	9.5	9.5	9.5	9.8	9.8	10.2	10.8	10.9	9.7	8.4		
	PrecAvg	1040	82	81	120	102	96	77	72	75	82	98	84	70		
(15) Precipitation	PrecMax	1392	115	125	187	153	161	114	109	142	137	171	115	114		
	PrecMin	856	48	49	71	58	42	40	33	15	29	51	31	31		
	PrecStd	117	14	19	29	23	28	19	23	27	24	26	20	21		
	PrecStd	117	14	19	29	23	28	19	23	27	24	26	20	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.9	6.9	5.4	4.5	3.8	2.8	2.7	3.3	4.6	6.3	6.6		
		MCWB	5.1	5.3	4.9	4.0	3.0	2.4	1.5	1.6	2.1	2.9	4.1	4.6		
	2%	DB	5.8	5.6	5.3	4.1	3.6	2.5	1.8	1.7	2.3	3.1	4.0	5.1		
		MCWB	4.2	4.2	4.0	3.0	2.4	1.4	0.9	0.8	1.5	2.0	2.7	3.5		
	5%	DB	5.0	4.9	4.6	3.3	2.8	1.6	1.2	1.0	1.7	2.4	3.2	4.1		
		MCWB	3.6	3.6	3.5	2.4	1.9	0.7	0.4	0.3	1.0	1.4	2.1	2.7		
	10%	DB	4.3	4.2	4.0	2.6	2.0	0.8	0.4	0.3	1.1	1.7	2.5	3.3		
		MCWB	3.0	3.0	3.0	1.8	1.3	0.1	-0.3	-0.2	0.5	0.9	1.6	2.0		
	0.4%	WB	5.6	5.5	5.2	4.2	3.5	2.6	1.9	1.9	2.6	3.3	4.4	5.0		
		MCDB	7.0	6.5	6.2	5.0	4.1	3.5	2.3	2.3	3.0	4.1	6.1	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	4.5	4.5	4.2	3.2	2.6	1.6	1.1	1.0	1.8	2.2	3.0	3.7		
		MCDB	5.5	5.3	5.0	3.9	3.3	2.3	1.6	1.4	2.1	2.9	3.7	4.7		
	5%	WB	3.8	3.8	3.6	2.6	2.0	0.8	0.5	0.4	1.1	1.6	2.3	2.9		
		MCDB	4.7	4.6	4.3	3.2	2.7	1.5	1.1	0.8	1.5	2.2	3.0	3.8		
	10%	WB	3.2	3.3	3.2	1.9	1.4	0.1	-0.2	-0.2	0.6	1.0	1.8	2.3		
		MCDB	4.1	4.0	3.9	2.6	2.0	0.8	0.4	0.3	1.0	1.6	2.4	3.0		
	5% DB	MDBR	2.7	2.7	3.1	3.5	3.9	4.6	5.2	4.7	4.4	3.3	3.0	2.7		
		MCDBR	4.0	3.4	3.4	3.5	4.1	4.5	4.5	4.3	3.8	3.7	4.1	3.8		
		MCWBR	2.9	2.7	2.9	3.2	3.6	4.1	4.3	4.1	3.6	3.1	3.4	2.9		
		MCWBR	3.7	3.0	3.2	3.3	3.9	4.4	4.3	4.3	3.7	3.5	3.9	3.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	2.9	2.6	3.0	3.2	3.5	4.2	4.3	4.2	3.7	3.1	3.3	2.9		
		MCWBR	2.9	2.6	3.0	3.2	3.5	4.2	4.3	4.2	3.7	3.1	3.3	2.9		
	taub	taub	0.308	0.308	0.297	0.279	0.223	0.366	0.209	0.267	0.287	0.309	0.304	0.310		
		taud	2.494	2.501	2.502	2.374	2.097	2.834	2.044	2.237	2.369	2.400	2.456	2.463		
(42) Clear-Sky Solar Irradiance	Ebn at Noon	Ebn at Noon	967	918	842	677	444	304	439	661	829	903	966	978		
		Edh at Noon	98	87	71	52	31	20	31	57	78	95	101	104		
	RadAvg	RadAvg	5.05	3.64	1.95	0.85	0.25	0.10	0.18	0.64	1.67	3.11	4.77	5.72		
(45) All-Sky Solar Radiation	RadStd	RadStd	0.46	0.28	0.16	0.07	0.02	0.00	0.01	0.05	0.13	0.21	0.29	0.46		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	-0.26	-0.42	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page